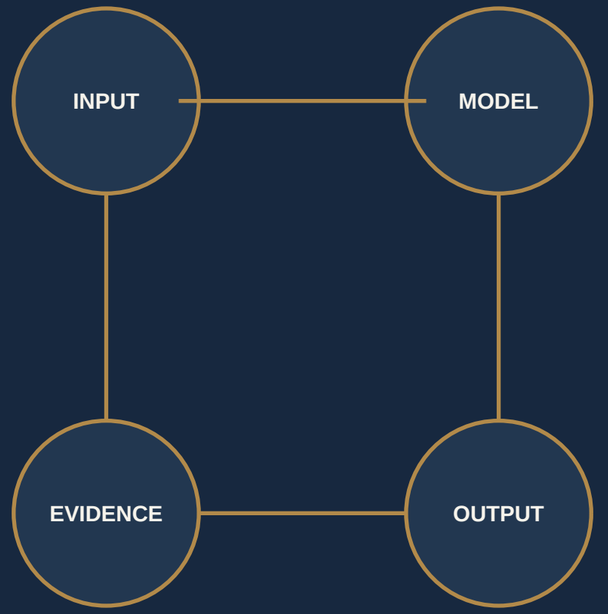
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Reinforcement Learning

AI.205 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define agents.
- Apply key methods including states.
- Build or analyze a practical system involving actions.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Agents

02

METHODS

States

03

SYSTEMS

Actions

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Agents

Agents

02 · States

States

03 · Actions

Actions

04 · Rewards

Rewards

05 · Policies

Policies

06 · Value functions

Value functions

07 · Markov decision processes

Markov decision processes

08 · Q-learning

Q-learning

Core ideas, made explicit

AI.205 • Distinguish definitions, assumptions and evidence

- Agents
- States
- Actions
- Rewards
- Policies

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.205 • Distinguish definitions, assumptions and evidence

- Value functions
- Markov decision processes
- Q-learning
- Policy learning
- Exploration and exploitation

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Agents

INPUTS
States

METHOD
Actions

OUTPUT
Rewards

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving agents.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using agents.
- Implement or simulate a baseline using states.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.